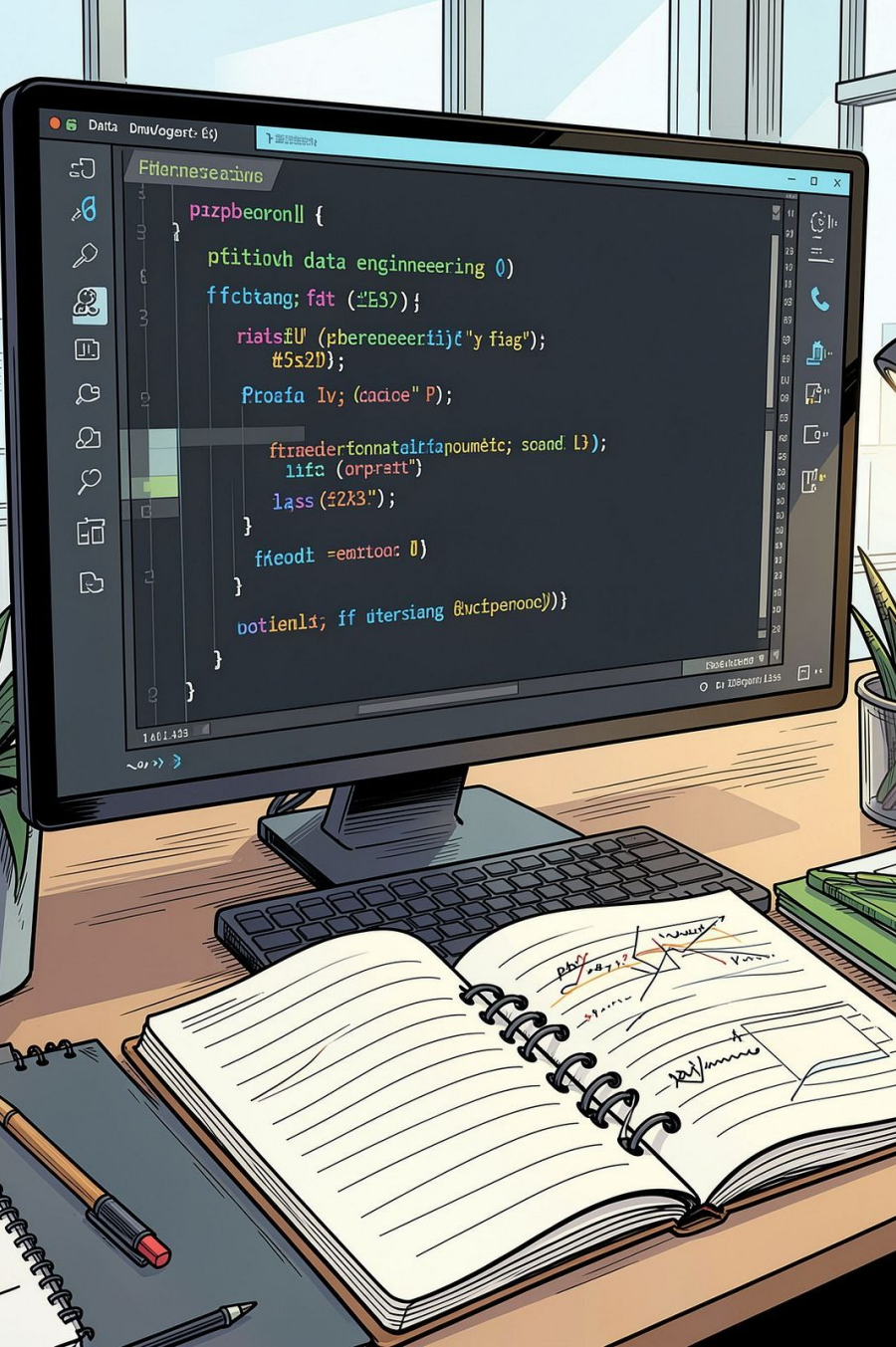


Customer Shopping Behaviour Analysis

Project documenting an end-to-end analytics workflow: data cleaning & feature engineering in Python (Pandas), analytical queries in MySQL, and interactive visualizations in Power BI. The dataset (shopping_behavior_updated.csv) contains 3,900 customer records and 16 columns capturing demographics, purchases, ratings, subscription status, discounts, payment methods and frequency. This deck summarizes methods, key transformations, representative SQL queries, dashboard design, and business insights.





Project Overview & Workflow

Stage 1 — Python (Data Cleaning)

Load CSV with pandas, inspect head(), run data.describe() and null checks, standardize column names, create derived features (age groups, purchase frequency in days), and prepare dataset for database upload.

Stage 2 — MySQL (Analysis)

Load cleaned table into MySQL and execute analytical queries to answer revenue, discount, product ranking, segmentation and cohort-style questions using aggregation and window functions.

Stage 3 — Power BI (Visualization)

Publish interactive dashboard with KPI cards, slicers (Subscription Status, Gender, Category, Location) and charts for revenue, sales counts, age-group comparisons and subscription behaviour.

Dataset Overview — Raw & Cleaned

Source file: shopping_behavior_updated.csv — 3,900 records, 16 columns: Customer ID, Age, Gender, Item Purchased, Category, Purchase Amount (USD), Location, Size, Color, Season, Review Rating, Subscription Status, Discount Applied, Previous Purchases, Payment Method, Frequency of Purchases.

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Discupice		Y9%
Pleasins		\$ 53ncan
MEBAN		\$970
FHDWICS		\$299
RKSEN		\$090
TOSOA		\$500
MELIZA		- 5%
MACIN		\$ 330
MUN IX		\$ 364
MAINIX		\$878
MLXIZAN		- 900
NIDIMIX		- 696
MANIX		- 5%

Statistical Snapshot

Customer count: 3,900.
Mean age: 44.07 (range 18–70).
Mean purchase amount: 59.76
USD (range 20–100). Mean
review rating: 3.75 (range 2.5–
5.0).



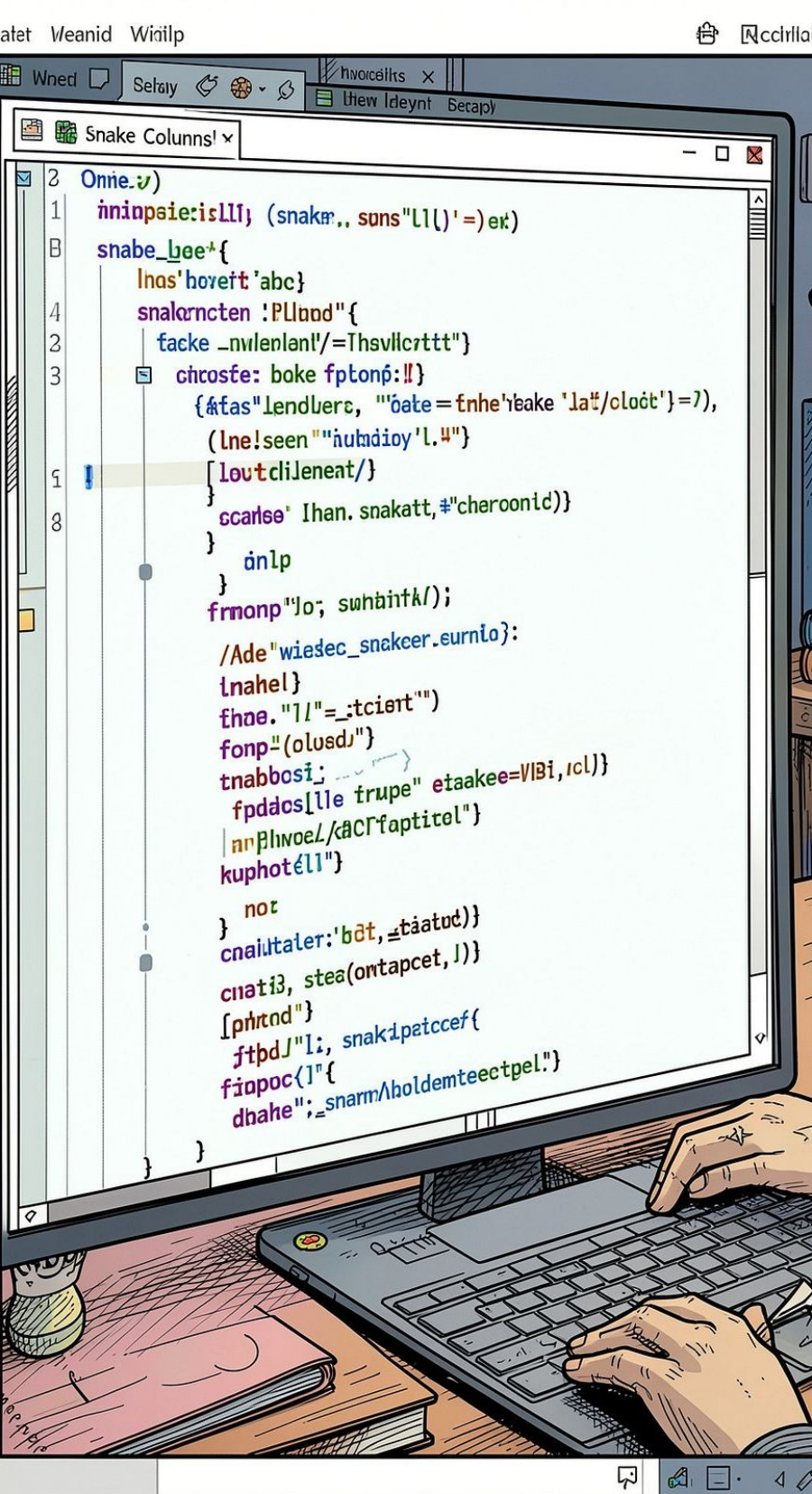
Data Quality

Null check:
data.isnull().sum() returned 0 for
all 16 fields — no missing values
detected prior to feature
engineering.



Categorical Diversity

Item Purchased: 25 unique values (top = Blouse, freq 171).
Category: 4 unique (top = Clothing, freq 1,737). Gender has 2 values (top = Male, freq 2,652).



Representative code and transformations used to standardize and enrich the dataset before database load. All operations executed in pandas.

Standardize Column Names

data.columns = data.columns.str.lower(); replace spaces with underscores; rename purchase_amount column to purchase_amount for consistency and SQL friendliness.

Age Grouping

Created quantile-based age_group with labels ['Young Adult', 'Adult', 'Middle-aged', 'Senior'] via pd.qcut to enable revenue and purchase comparisons across life stages.

Purchase Frequency → Days

Mapped textual frequency to numeric days using mapping: Weekly=7, Fortnightly/Bi-Weekly=14, Monthly=30, Quarterly/Every 3 Months=90, Annually=365. Result column: purchase_frequency_days.

Final cleaned DataFrame columns (sample):

['customer_id', 'age', 'gender', 'item_purchased', 'category', 'purchase_amount', 'location', 'size', 'color', 'season', 'review_rating', 'subscription_status', 'discount_applied', 'previous_purchases', 'payment_method', 'frequency_of_purchases', 'age_group', 'purchase_frequency_days'].

Load to MySQL — ETL Summary

Cleaned DataFrame written to MySQL using SQLAlchemy to a database named customer_behaviour and table customer:

```
from sqlalchemy import create_engine
```

```
username = "root"
```

```
password = "*****" # replace as needed
```

```
engine = create_engine(f"mysql+pymysql://{username}:  
{password}@localhost:3306/customer_behaviour")
```

```
data.to_sql("customer", engine, if_exists="replace", index=False)
```

📌 Data successfully loaded into table 'customer' in database 'customer_behaviour'.

Once in MySQL, the dataset is ready for aggregation, segmentation, window functions and downstream reporting in Power BI.



Representative MySQL Queries & Business Purpose



Revenue by Gender

```
SELECT gender,  
SUM(purchase_amount)  
AS revenue FROM  
customer GROUP BY  
gender; Purpose:  
Compare total revenue  
contributions by male vs.  
female customers.
```



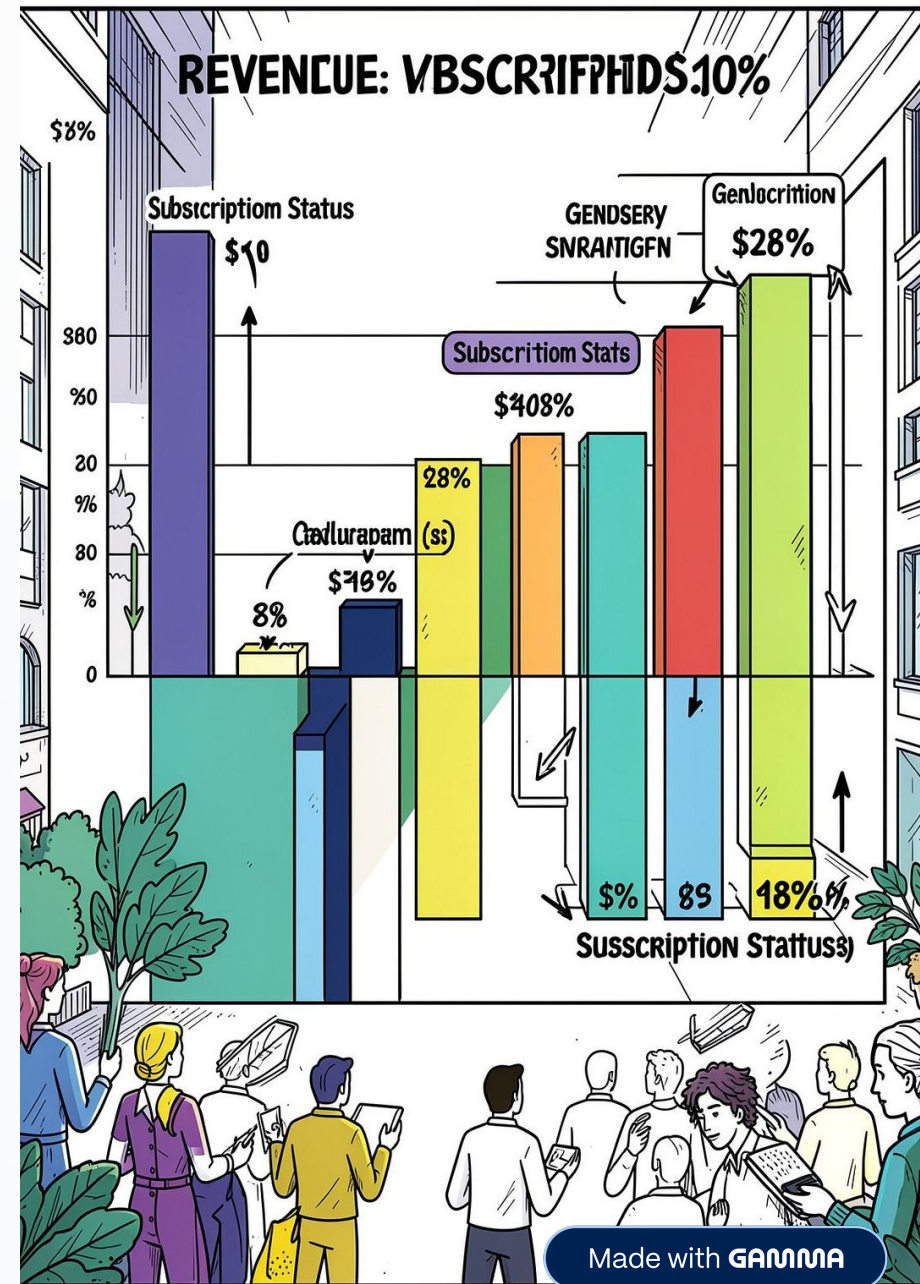
High-Spending Customers Despite Discounts

```
SELECT  
customer_id,  
purchase_amount FROM  
customer WHERE  
discount_applied='Yes'  
AND purchase_amount  
>= (SELECT  
AVG(purchase_amount)  
FROM customer);  
Purpose: Identify  
customers who still spend  
at/above average even  
when using discounts.
```



Top Products by Average Rating

```
SELECT item_purchased,  
AVG(review_rating) AS  
Average_Product_Rating  
FROM customer GROUP  
BY item_purchased  
ORDER BY  
AVG(review_rating) DESC  
LIMIT 5; Purpose: Surface  
highest-rated products  
for merchandising or  
promotion.
```



Advanced MySQL Queries — Segmentation & Rankings

Subscribers vs. Non-Subscribers Spend

Aggregation: subscription_status, total_customer, avg_spend, total_revenue. Purpose: compare counts, average purchase and revenue by subscription status to inform retention and acquisition strategy.

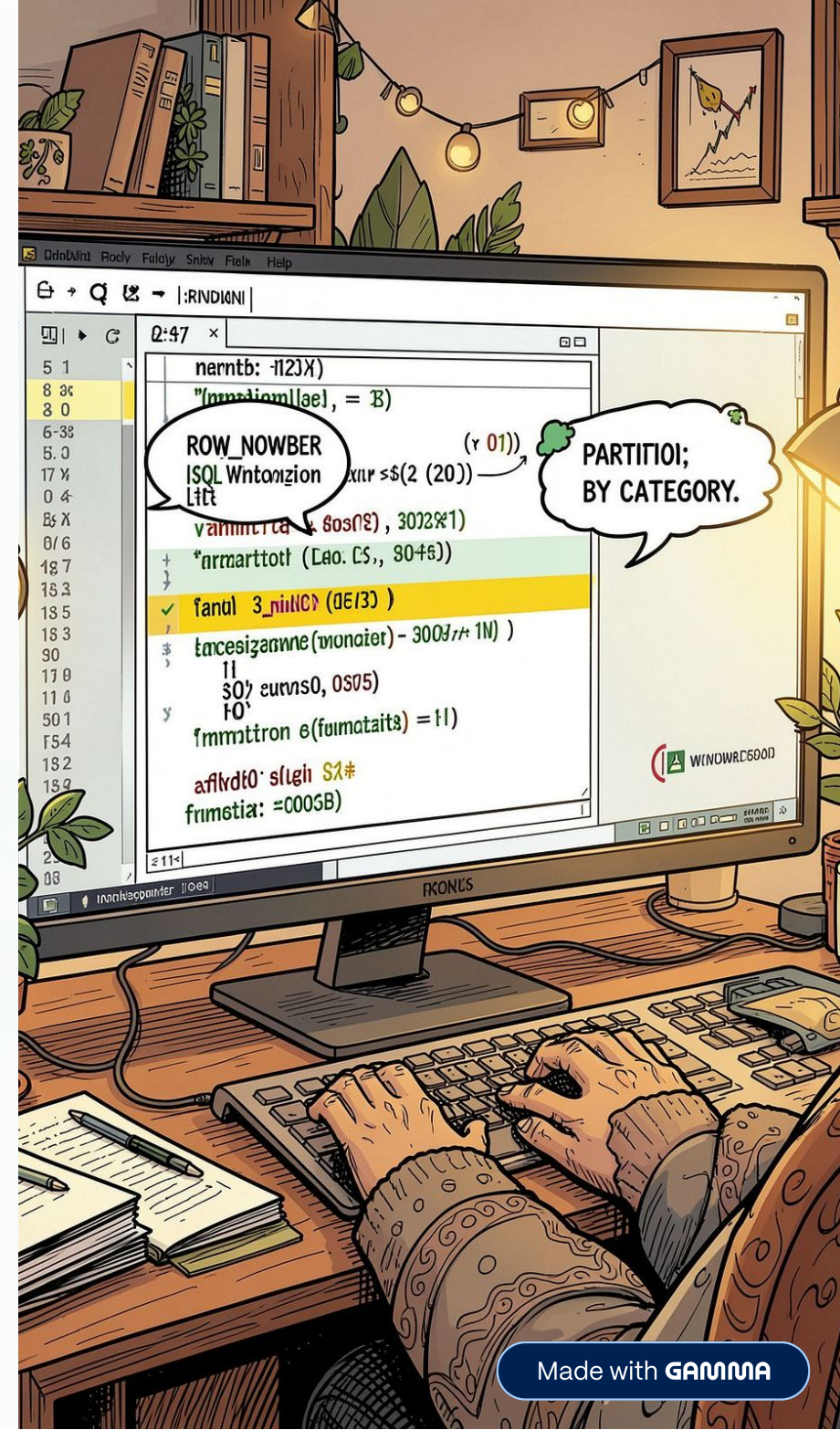
Top 3 Products per Category

Use ROW_NUMBER() OVER (PARTITION BY category ORDER BY COUNT(*) DESC) to rank products and return top 3 per category—actionable for category-level assortment and stocking.

Customer Segmentation by Purchase History

Classify as new (previous_purchases=1), returning (2–10), loyal (>10) and count each segment. Purpose: tailor marketing and loyalty programs to lifecycle stage.

Another useful query computes repeat buyers vs. subscription rate (repeat threshold >5 purchases) to test whether higher-frequency buyers are more likely to subscribe.



Power BI — Dashboard Design & Key Visuals

The dashboard consolidates cleaned data into interactive visuals with slicers for Subscription Status, Gender, Category and Location. KPI cards and charts provide both top-line and diagnostic insights.



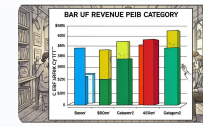
KPI Cards

Average Purchase Amount: 59.76 USD. Average Review Rating: 3.75. Number of Customers: ~4,000.



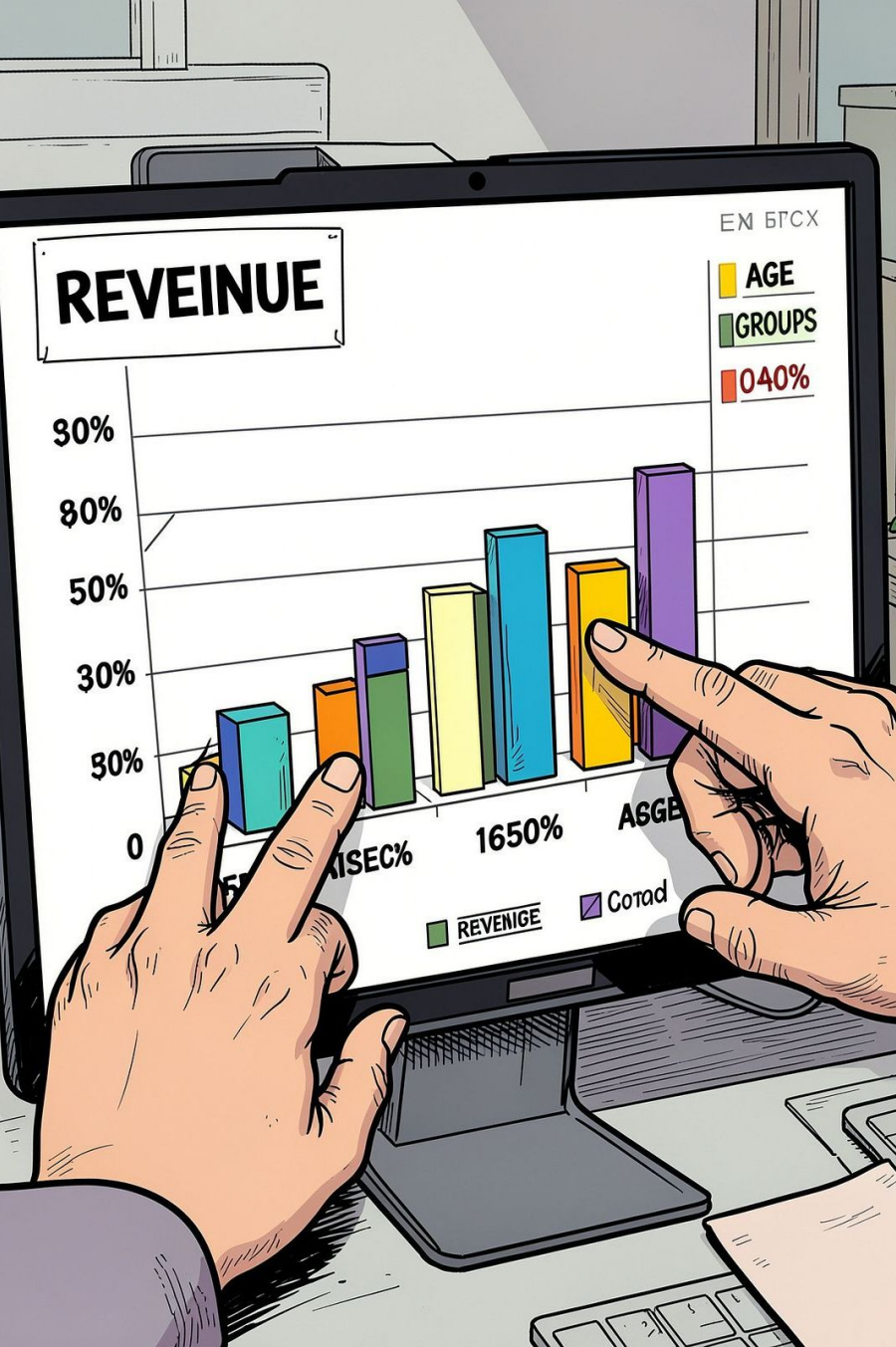
Subscriber Distribution

% of Customers by Subscription Status — Donut: No 73%, Yes 27%. Used as a primary slicer and retention KPI.



Revenue & Sales by Category

Revenue by Category (sum of purchase_amount) and Sales by Category (purchase counts) surface top-performing categories for assortment and promotions.



Age Group & Repeat Buyer Insights

Age groups were created using quantile bins (Young Adult, Adult, Middle-aged, Senior) to allow comparable cohort analysis. Key queries produce total revenue and percent contribution by age group, and sales counts by age group to identify which life stages drive the most revenue vs. transaction volume.



Revenue Contribution by Age Group

SQL aggregates `SUM(purchase_amount)` grouped by computed age buckets and computes percent of total revenue. This identifies which cohorts to prioritize for targeted offers.



Repeat Buyers & Subscription Rate

Query groups customers with `previous_purchases > 5` as 'Repeat Buyers', compares subscription counts and subscription rate (%) to test correlation between loyalty and subscription adoption.

Summary, Recommendations & Next Steps

Summary: The project demonstrates a repeatable pipeline — CSV ingestion, robust cleaning and feature engineering in Python, analytical querying in MySQL (aggregations, window functions, segmentation) and an interactive Power BI dashboard surfaced through KPI cards and slicers. Key metrics: average purchase amount 59.76 USD; average review rating 3.75; ~3,900 customers in dataset. Subscriber split: ~27% subscribed, 73% not subscribed (donut metric).

01	02	03
Short-Term Actions	Medium-Term	Long-Term
Use SQL results to target high-spending discount users with loyalty offers; promote top-rated products; tailor promotions by age group and category using dashboard slicers.	A/B test subscription incentives for repeat buyers; enrich dataset with visit/channel data to improve attribution and personalization.	Operationalize pipeline: scheduled ETL, maintain MySQL analytical tables, automate Power BI refresh, and integrate predictive models (CLV, churn risk) into the dashboard.

📄 Deliverable: reproducible documentation and an interactive Power BI report that enables stakeholders to explore revenue, ratings, discounts, subscription behaviour, category performance, and cohort insights across age segments.